

E-Commerce Logistics: EDA Mini-Project

This project performs EDA on an e-commerce retail dataset to identify sales trends, customer purchasing behavior, product performance, and revenue patterns using Python.

Source link: <https://amanxai.com/2025/03/02/try-these-datasets-to-master-data-cleaning/>

1. importing libraries and loading the dataset

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns

df = pd.read_csv("data.csv", encoding='latin1')

df.head(5)

{"type": "dataframe", "variable_name": "df"}
```

2. initial data exploration

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   InvoiceNo        541909 non-null object  
 1   StockCode       541909 non-null object  
 2   Description     540455 non-null object  
 3   Quantity        541909 non-null int64   
 4   InvoiceDate      541909 non-null object  
 5   UnitPrice       541909 non-null float64  
 6   CustomerID      406829 non-null float64  
 7   Country         541909 non-null object  
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB

df.describe()

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"Quantity\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 196412.42266088672,\n        \"min\": -80995.0,\n        \"max\": 541909.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          9.55224954743324,\n          3.0,\n          541909.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"UnitPrice\",
```

```

{"properties": {"dtype": "number", "std": 190752.07570771928, "min": -11062.06, "max": 541909.0, "num_unique_values": 8, "samples": [4.611113626088513, 2.08, 541909.0]}, "semantic_type": "", "description": "", "column": "CustomerID", "properties": {"dtype": "number", "std": 139204.16800694188, "min": 1713.6003033216632, "max": 406829.0, "num_unique_values": 8, "samples": [15287.690570239585, 15152.0, 406829.0]}, "semantic_type": "", "description": "", "type": "dataframe"}

```

```
df.isnull().sum().sort_values(ascending=False)
```

```

CustomerID      135080
Description      1454
StockCode         0
InvoiceNo        0
Quantity         0
InvoiceDate      0
UnitPrice        0
Country          0
dtype: int64

```

3. data cleaning and preprocessing

```

duplicates = df.duplicated().sum()

print(f"Duplicate rows: {duplicates}")

Duplicate rows: 5268

df = df.drop_duplicates()

print(f"Updated shape: {df.shape}")

Updated shape: (536641, 8)

negative_quantity = df[df['Quantity'] < 0]

print(f"Transactions with negative quantity:
{negative_quantity.shape[0]}")

Transactions with negative quantity: 10587

negative_price = df[df['UnitPrice'] < 0]

print(f"Transactions with negative unit price:
{negative_price.shape[0]}")

```

Transactions with negative unit price: 2

Handling Invalid Transactions-

Negative quantities may represent cancelled or returned orders.

Negative unit prices may indicate pricing adjustments or data entry errors.

These records will be removed to focus the analysis on valid completed transactions.

```
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]
print(f"Cleaned dataset shape: {df.shape}")
Cleaned dataset shape: (524878, 8)
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
print(df['InvoiceDate'].dtype)
datetime64[ns]
# Feature Engineering- TotalPrice column
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']
df.head(3)
{"type": "dataframe", "variable_name": "df"}
```

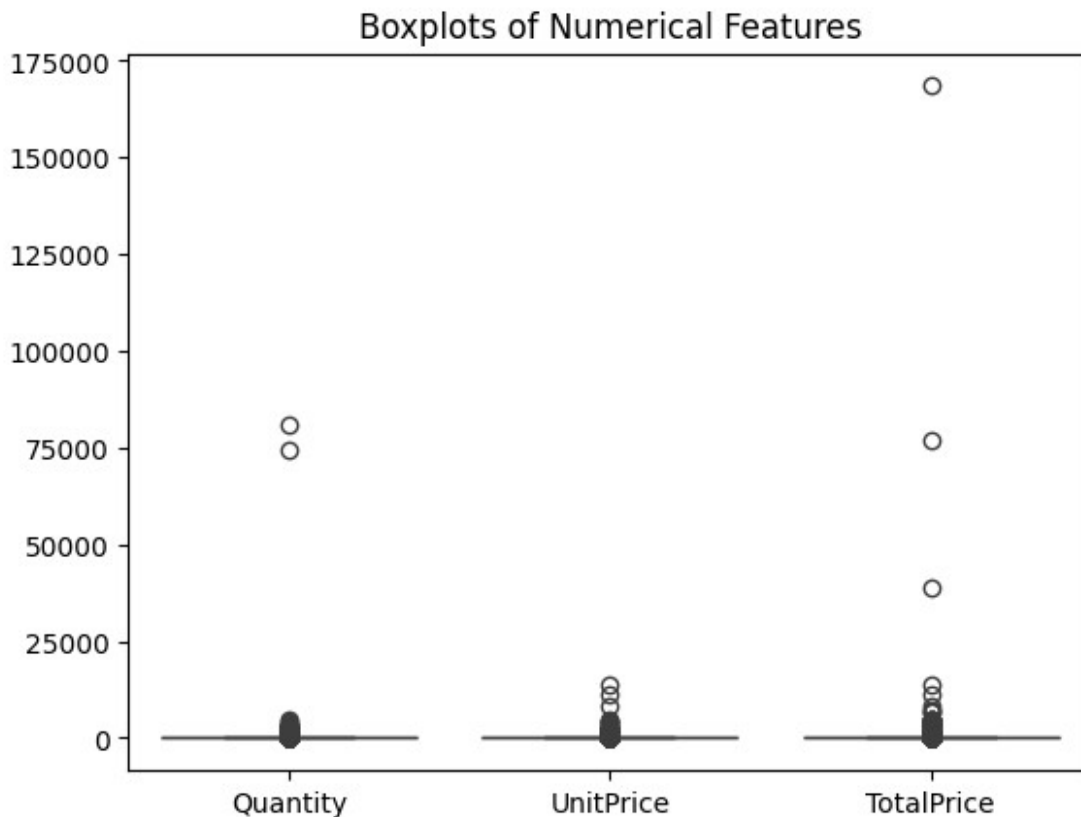
4. EDA

```
df[['Quantity', 'UnitPrice', 'TotalPrice']].describe()
{"summary": "{\n  \"name\": \"df[['Quantity', 'UnitPrice', 'TotalPrice']]\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"Quantity\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 183670.51608009043,\n        \"min\": 1.0,\n        \"max\": 524878.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          524878.0,\n          10.616600429052085,\n          11.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"UnitPrice\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 184946.69931668125,\n        \"min\": 0.001,\n        \"max\": 524878.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          3.9225725673394587,\n          2.08,\n          524878.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"TotalPrice\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 186597.94365421467,\n        \"min\": 0.001,\n        \"max\": 524878.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n
```

```
20.27539886221179,\n          9.92,\n          524878.0\n\n\"semantic_type\": \"\", \n\n\"description\": \"\" \n\n}\n ]\n}","type":"dataframe"}

# checking for any outliers.
sns.boxplot(data=df[['Quantity', 'UnitPrice', 'TotalPrice']])

plt.title("Boxplots of Numerical Features")
plt.show()
```



Outlier Observation in Quantity, UnitPrice, and TotalPrice

The dataset contains several extreme values in Quantity, UnitPrice, and TotalPrice.

These may represent:

- bulk purchases,
- wholesale transactions,
- premium products,
- or unusually large orders.

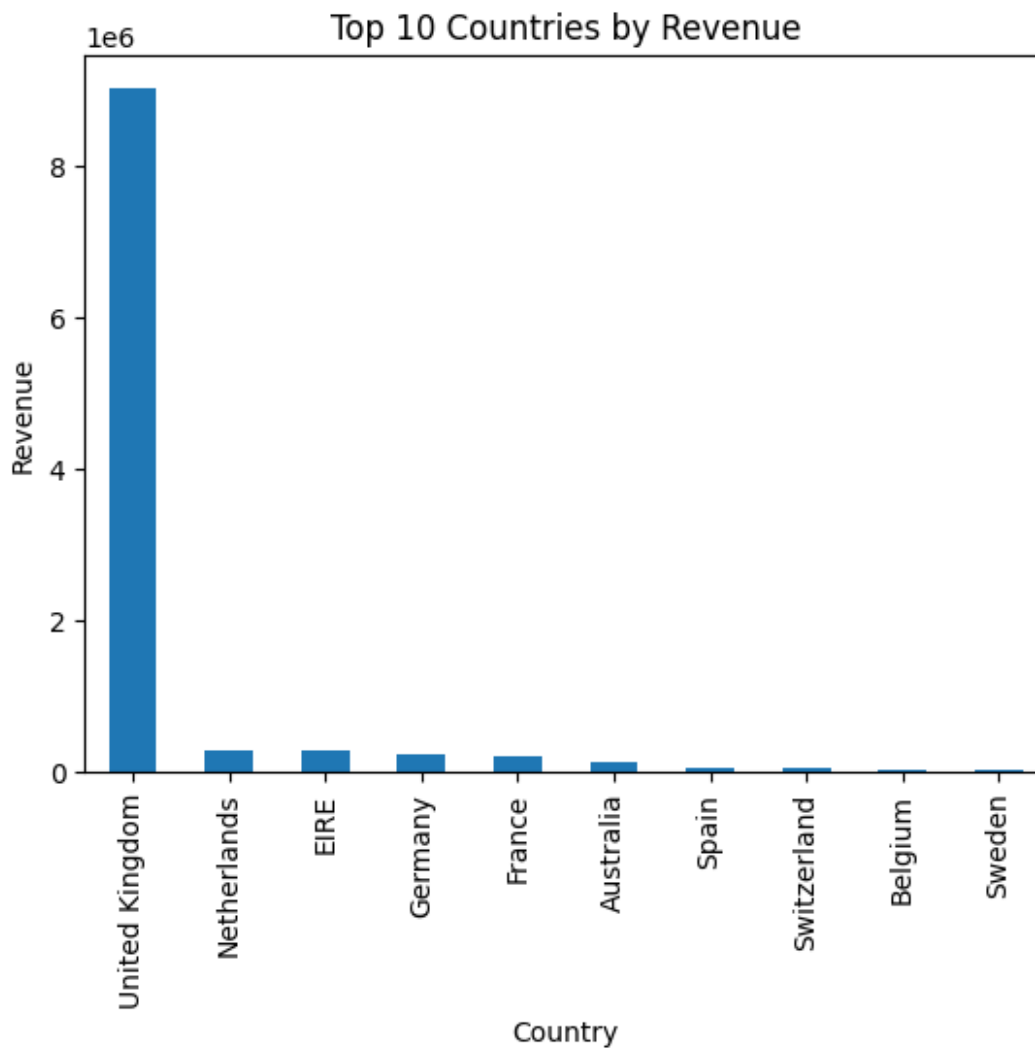
Since these transactions may contain valuable business information, the outliers were retained for analysis.

```
country_revenue = (  
    df.groupby('Country')['TotalPrice']  
    .sum()  
    .sort_values(ascending=False)  
    .head(10)  
)
```

country_revenue

```
Country  
United Kingdom    9001744.094  
Netherlands       285446.340  
EIRE              283140.520  
Germany           228678.400  
France            209625.370  
Australia         138453.810  
Spain             61558.560  
Switzerland       57067.600  
Belgium           41196.340  
Sweden            38367.830  
Name: TotalPrice, dtype: float64
```

```
country_revenue.plot(kind='bar')  
plt.title("Top 10 Countries by Revenue")  
plt.xlabel("Country")  
plt.ylabel("Revenue")  
plt.show()
```



```
top_products = (
    df.groupby('Description')['Quantity']
      .sum()
      .sort_values(ascending=False)
      .head(10)
)
```

top_products

Description	
PAPER CRAFT , LITTLE BIRDIE	80995
MEDIUM CERAMIC TOP STORAGE JAR	78033
WORLD WAR 2 GLIDERS ASSTD DESIGNS	54951
JUMBO BAG RED RETROSPOT	48371
WHITE HANGING HEART T-LIGHT HOLDER	37872
POPCORN HOLDER	36749
PACK OF 72 RETROSPOT CAKE CASES	36396
ASSORTED COLOUR BIRD ORNAMENT	36362

```
RABBIT NIGHT LIGHT          30739
MINI PAINT SET VINTAGE      26633
Name: Quantity, dtype: int64
```

```
# Inspect unusually high quantity transaction
```

```
df[df['Description'] == 'PAPER CRAFT , LITTLE BIRDIE']
```

```
{"summary":{"\n  \"name\": \"df[df['Description'] == 'PAPER CRAFT ,  
LITTLE BIRDIE']\",\n  \"rows\": 1,\n  \"fields\": {\n    \"column\": \"InvoiceNo\",\n    \"properties\": {\n      \"dtype\": \"string\",\n      \"num_unique_values\": 1,\n      \"samples\": [\n        \"581483\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"StockCode\",\n    \"properties\": {\n      \"dtype\": \"string\",\n      \"num_unique_values\": 1,\n      \"samples\": [\n        \"23843\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"Description\",\n    \"properties\": {\n      \"dtype\": \"string\",\n      \"num_unique_values\": 1,\n      \"samples\": [\n        \"PAPER CRAFT , LITTLE BIRDIE\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"Quantity\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": null,\n      \"min\": 80995,\n      \"max\": 80995,\n      \"num_unique_values\": 1,\n      \"samples\": [\n        80995\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"InvoiceDate\",\n    \"properties\": {\n      \"dtype\": \"date\",\n      \"min\": \"2011-12-09 09:15:00\",\n      \"max\": \"2011-12-09 09:15:00\",\n      \"num_unique_values\": 1,\n      \"samples\": [\n        \"2011-12-09 09:15:00\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"UnitPrice\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": null,\n      \"min\": 2.08,\n      \"max\": 2.08,\n      \"num_unique_values\": 1,\n      \"samples\": [\n        2.08\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"CustomerID\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": null,\n      \"min\": 16446.0,\n      \"max\": 16446.0,\n      \"num_unique_values\": 1,\n      \"samples\": [\n        16446.0\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"Country\",\n    \"properties\": {\n      \"dtype\": \"string\",\n      \"num_unique_values\": 1,\n      \"samples\": [\n        \"United Kingdom\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"TotalPrice\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": null,\n      \"min\": 168469.6,\n      \"max\": 168469.6,\n
```

```
\n      num_unique_values\": 1,\n      samples\": [\n        168469.6\n      ],\n      semantic_type\": \"\", \n      description\": \"\" \n    } \n  ] \n}\", "type": "dataframe"}
```

Observation

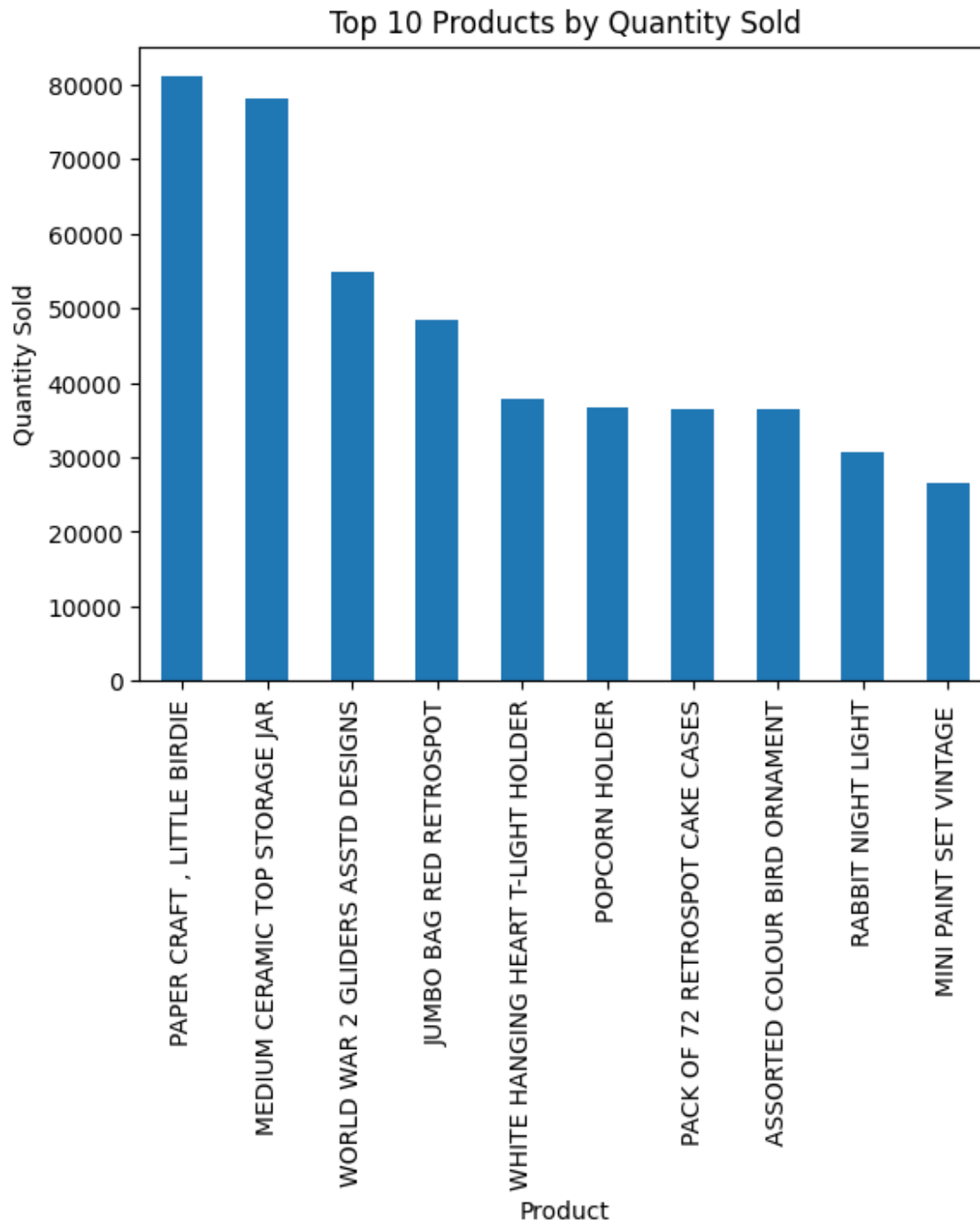
The product "PAPER CRAFT , LITTLE BIRDIE" shows an unusually high purchase quantity of 80,995 units in a single transaction.

This may indicate:

- a bulk wholesale order,
- inventory movement,
- or an exceptional commercial purchase.

The transaction was retained because it represents valid business activity rather than a data error.

```
top_products.plot(kind='bar')
plt.title("Top 10 Products by Quantity Sold")
plt.xlabel("Product")
plt.ylabel("Quantity Sold")
plt.show()
```



```
df['YearMonth'] = df['InvoiceDate'].dt.to_period('M')  
monthly_revenue = (  
    df.groupby('YearMonth')['TotalPrice']  
    .sum()  
)
```

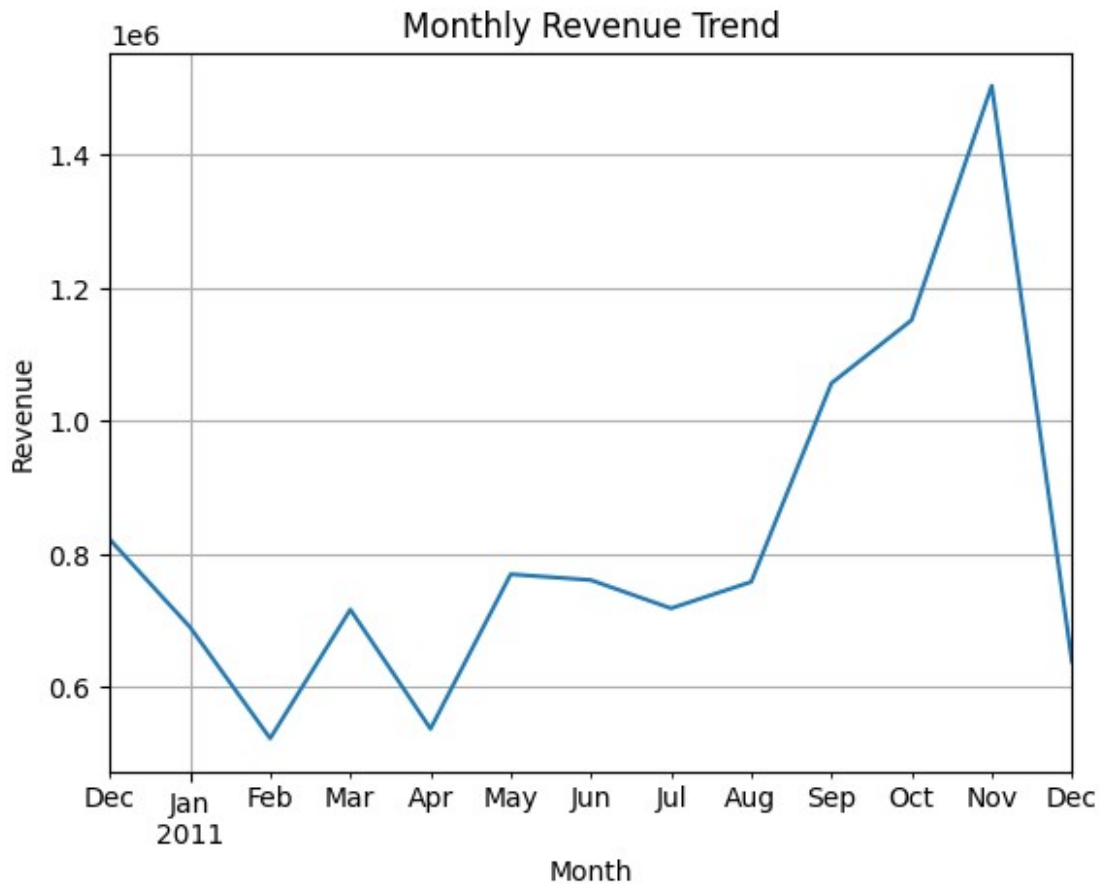
```
monthly_revenue
```

```
YearMonth
```

2010-12	821452.730
2011-01	689811.610
2011-02	522545.560
2011-03	716215.260
2011-04	536968.491
2011-05	769296.610
2011-06	760547.010
2011-07	718076.121
2011-08	757841.380
2011-09	1056435.192
2011-10	1151263.730
2011-11	1503866.780
2011-12	637790.330

```
Freq: M, Name: TotalPrice, dtype: float64
```

```
monthly_revenue.plot()  
plt.title("Monthly Revenue Trend")  
plt.xlabel("Month")  
plt.ylabel("Revenue")  
plt.grid(True)  
plt.show()
```



Monthly Revenue Growth and Sharp Decline-

Revenue shows significant growth during September–November 2011, with November recording the highest sales.

The sharp decline in December 2011 is perhaps due to incomplete monthly data rather than an actual drop in business performance.

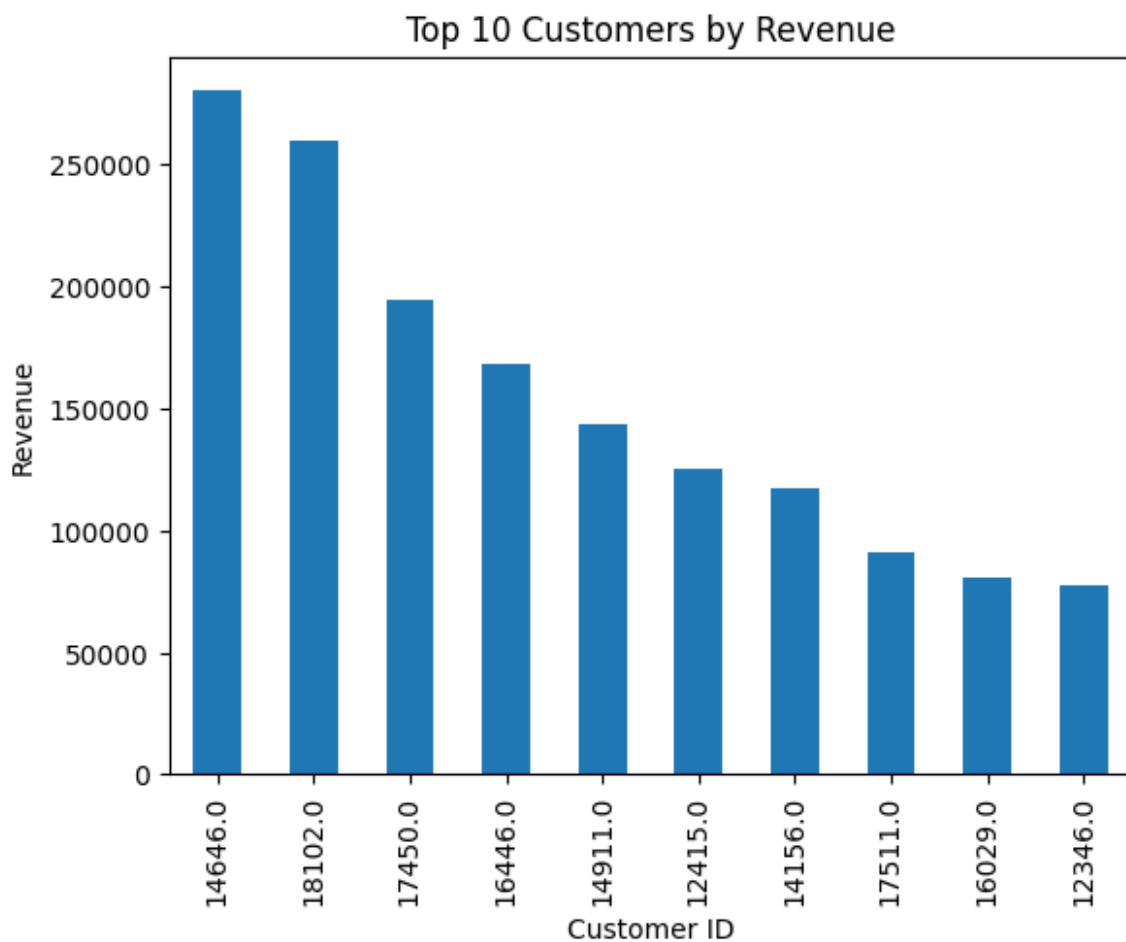
```
top_customers = (
    df.groupby('CustomerID')['TotalPrice']
      .sum()
      .sort_values(ascending=False)
      .head(10)
)
```

top_customers

CustomerID	TotalPrice
14646.0	280206.02
18102.0	259657.30
17450.0	194390.79
16446.0	168472.50

```
14911.0    143711.17
12415.0    124914.53
14156.0    117210.08
17511.0     91062.38
16029.0     80850.84
12346.0     77183.60
Name: TotalPrice, dtype: float64
```

```
top_customers.plot(kind='bar')
plt.title("Top 10 Customers by Revenue")
plt.xlabel("Customer ID")
plt.ylabel("Revenue")
plt.show()
```



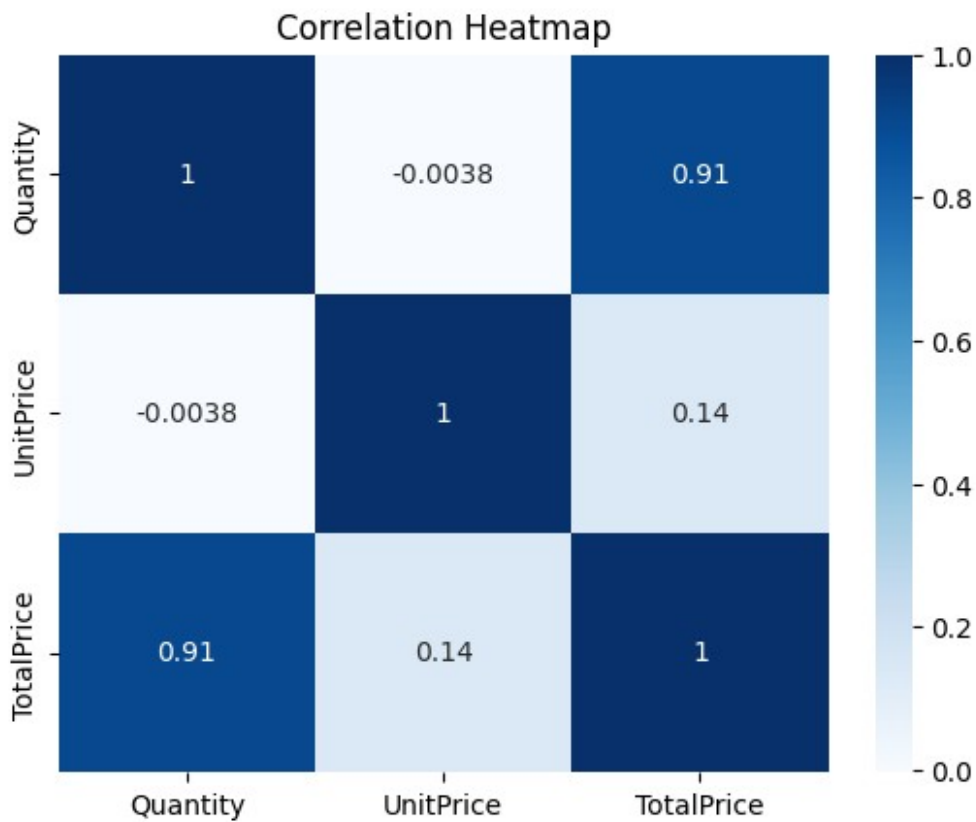
```
correlation_matrix = df[['Quantity', 'UnitPrice',  
                        'TotalPrice']].corr()  
  
correlation_matrix
```

```

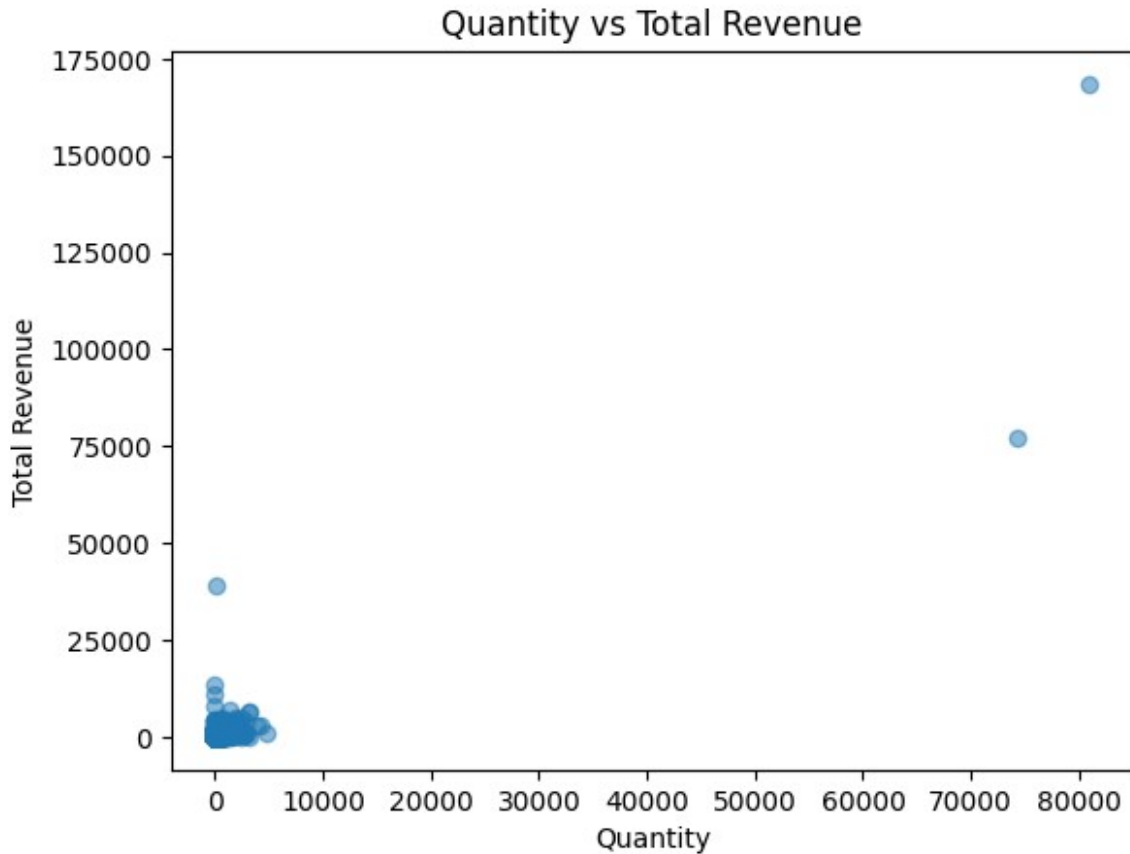
{"summary":{"\n  \"name\": \"correlation_matrix\",\n  \"rows\": 3,\n  \"fields\": [\n    {\n      \"column\": \"Quantity\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0.5547421835078391,\n        \"min\": -0.0037884185655402535,\n        \"max\": 1.0,\n        \"num_unique_values\": 3,\n        \"samples\": [\n          1.0,\n          -0.0037884185655402535,\n          0.9074018676033929\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"UnitPrice\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0.5433892807966343,\n        \"min\": -0.0037884185655402535,\n        \"max\": 1.0,\n        \"num_unique_values\": 3,\n        \"samples\": [\n          -0.0037884185655402535,\n          1.0,\n          0.13738104867793038\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"TotalPrice\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0.4735711722085651,\n        \"min\": 0.13738104867793038,\n        \"max\": 1.0,\n        \"num_unique_values\": 3,\n        \"samples\": [\n          0.9074018676033929,\n          0.13738104867793038,\n          1.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    ]\n  ]\n}, \"type\": \"dataframe\", \"variable_name\": \"correlation_matrix\"}

sns.heatmap(correlation_matrix, annot=True, cmap='Blues')
plt.title("Correlation Heatmap")
plt.show()

```



```
plt.scatter(  
    df['Quantity'],  
    df['TotalPrice'],  
    alpha=0.5  
)  
  
plt.title("Quantity vs Total Revenue")  
plt.xlabel("Quantity")  
plt.ylabel("Total Revenue")  
  
plt.show()
```



Correlation Insights

- Quantity and TotalPrice show a strong positive correlation, indicating that larger purchase volumes significantly increase revenue.
- UnitPrice has a weak correlation with Quantity, suggesting that product pricing does not strongly influence purchase volume in this dataset.
- TotalPrice is more heavily driven by quantity purchased than by unit price alone.
- Most transactions are concentrated at lower quantity and revenue levels, while a few extreme bulk orders contribute disproportionately high revenue.

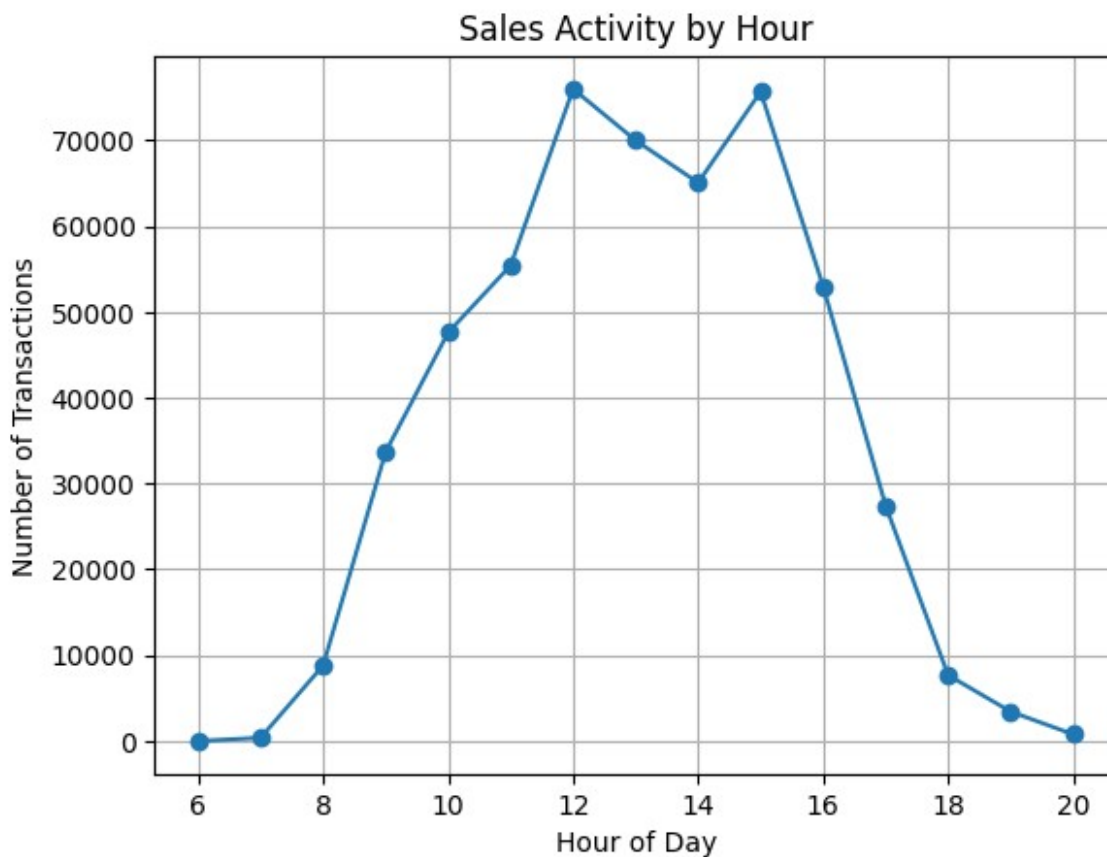
```
df['Hour'] = df['InvoiceDate'].dt.hour  
hourly_sales = df.groupby('Hour')['InvoiceNo'].count()
```

hourly_sales

Hour	
6	1
7	379
8	8797
9	33684
10	47597
11	55420
12	75986
13	69993

```
14    65057
15    75665
16    52992
17    27426
18     7676
19     3427
20      778
Name: InvoiceNo, dtype: int64

hourly_sales.plot(kind='line', marker='o')
plt.title("Sales Activity by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Number of Transactions")
plt.grid(True)
plt.show()
```



Customer Purchasing Behavior

Transaction activity peaks at 12 PM and 3 PM.

This suggests that customers are most active during standard business hours, with activity declining significantly during evening hours.

5. key findings

1. The United Kingdom generated the highest revenue by a significant margin, indicating a strong concentration of customers in that market.
2. Revenue showed strong growth during September–November 2011, with November recording the highest sales performance.
3. Several transactions contained unusually large purchase quantities, suggesting the presence of wholesale or bulk orders.
4. Quantity purchased showed a strong positive relationship with total revenue, making sales volume a key revenue driver.
5. Customer purchasing activity peaked during midday business hours, especially between 12 PM and 3 PM.
6. Some products demonstrated exceptionally high sales volumes, indicating strong product demand and popularity.
7. Negative quantities and unit prices were identified as cancelled transactions or pricing adjustments and were removed during data cleaning.